

ML Experiment No 07 - Comparission of Classification Algorithms

Import Libraries

```
In [5]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
from sklearn.svm import SVR
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

Load Sample Dataset

```
In [6]: from sklearn.datasets import fetch_california_housing
data = fetch_california_housing()
X, y = data.data, data.target
```

```
In [7]: X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2, random_state = 42)
```

```
In [9]: scaler_X = StandardScaler()
scaler_y = StandardScaler()
X_train_scaled = scaler_X.fit_transform(X_train)
X_test_scaled = scaler_X.transform(X_test)
y_train_scaled = scaler_y.fit_transform(y_train.reshape(-1, 1)).ravel()
```

```
In [10]: models = {
    "Linear Regression": LinearRegression(),
    "Polynomial Regression": LinearRegression(),
    "Decision Tree": DecisionTreeRegressor(random_state=42),
    "Random Forest": RandomForestRegressor(n_estimators=100, random_state=42),
```

```
"SVR": SVR(kernel='rbf')
}
```

```
In [11]: poly = PolynomialFeatures(degree=2)
X_poly_train = poly.fit_transform(X_train)
X_poly_test = poly.transform(X_test)
```

```
In [20]: results = {}

for name, model in models.items():
    if name == "Polynomial Regression":
        model.fit(X_poly_train, y_train)
        y_pred = model.predict(X_poly_test)

    elif name == "SVR":
        model.fit(X_train_scaled, y_train_scaled)
        y_pred = scaler_y.inverse_transform(
            model.predict(X_test_scaled).reshape(-1, 1)
        ).ravel()

    else:
        model.fit(X_train, y_train)
        y_pred = model.predict(X_test)

    results[name] = {
        "MAE": mean_absolute_error(y_test, y_pred),
        "MSE": mean_squared_error(y_test, y_pred),
        "R2_Score": r2_score(y_test, y_pred)
    }
```

Results

```
In [21]: results_df = pd.DataFrame(results).T
print(results_df)
```

	MAE	MSE	R2_Score
Linear Regression	0.533200	0.555892	0.575788
Polynomial Regression	0.467001	0.464301	0.645682
Decision Tree	0.454679	0.495235	0.622076
Random Forest	0.327543	0.255368	0.805123
SVR	0.397283	0.354151	0.729740

In []: